

ASSET MANAGEMENT SYSTEM

User and Administrator Manual

Django 5.2 LTS | HTMX-style partial updates | MySQL | Docker

Version 1.0

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Prepared from the supplied “Asset Identification Fields and Layout” requirements.

For internal organisational use

Document control

Document	Asset Management System - User and Administrator Manual
Application version	1.0 starter implementation
Audience	Asset staff, Stores, Divisions, Security, Approvers, Auditors and IT administrators
Database	MySQL in Docker; SQLite is used only for local development/testing when MySQL variables are absent
Deployment target	Windows Server 2022 with IIS reverse proxy, or another Docker-capable host
Source requirement	Asset Identification Fields and Layout.docx

Important: This package is a working, testable starter implementation. Before production use, complete organisational configuration, user-acceptance testing, security review, data-cleansing, backup testing and approval of workflow rules.

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1. System overview

The application is a modular Django monolith designed for internal asset-management workflows. It keeps the current asset state on the Asset record and records every transfer, return, gate movement and disposal action in transaction tables so the history remains auditable.

1.1 Included functions

- Asset identification and QR token generation.
- Procurement, invoice, supplier and acquisition-value details.
- Stock-entry, installation, warranty and operational-status details.
- Division, section, sub-section, room, location, employee and custodian records.
- Controlled transfers and returns with voucher history.
- Temporary and permanent gate-pass requests, Stores approval, Security outward/inward actions and overdue-return reporting.
- Staged Excel upload with validation preview and confirmation.
- Disposal, auction, write-off and passout records.
- Dashboard, audit events, CSV asset-register export and QR-based asset webpage.
- Role-based access using Django users, groups and permissions.

1.2 Main workflow

Step	Action	Expected result
1	Register or import the asset and procurement details.	A unique Asset Code, Transaction ID and QR token are stored.
2	Record installation, location and custodian.	The asset appears in the register with its current state.
3	Use Transfer or Return when responsibility changes.	A movement row is created and the Asset current state changes atomically.
4	Use Gate Pass when the asset crosses the security gate.	Stores and Security actions are recorded, with temporary returns tracked.
5	Use Disposal when the asset reaches end of life.	Auction, write-off and passout details are linked to the asset.
6	Use reports and audit history for review.	Authorised users can trace status, location and transaction history.

2. Roles and access

Access is controlled by Django permissions. The bootstrap command creates the following groups. The System Administrator should assign each user only the minimum role required for that person's work.

Role	Primary responsibility
System Administrator	Full configuration, user administration and technical oversight.
Asset Administrator	Registers and edits assets, procurement, installation, imports and disposal records.
Stores Officer	Approves transfers/returns and marks gate-pass requests after stores verification.
Division Officer	Views division assets and raises transfer, return and gate-pass requests.
Approving Officer	Approves controlled movement and disposal/write-off processes.
Security Officer	Uses gate-pass and QR screens to mark outward and inward movement.
Auditor	Read-only access to asset history, gate passes, disposal and audit events.
Report Viewer	Read-only access to operational reports and exports.

Security rule: Do not share accounts. Each approval and movement is stamped with the signed-in user. Shared accounts weaken the audit trail.

3. Signing in and navigation

3.1 Sign in

Step	Action	Expected result
1	Open the application address supplied by IT, for example https://asset-system.example.local .	The sign-in page opens.
2	Enter your username and password.	The dashboard opens if the account is active.
3	At first sign-in, change the initial password using the password-change page or ask the administrator.	The account no longer uses the deployment password.

3.2 Main menu

Assets: Search, register, edit and view QR-linked asset records.

Movements: Review transfer and return history; authorised users can execute new movements.

Gate Passes: Create requests, approve in Stores, and mark outward/inward in Security.

Disposal: Manage disposal, auction, write-off and passout details.

Reports: View status/location summaries, overdue temporary gate passes and download CSV.

Import: Upload and validate Excel workbooks.

Administration: Maintain users, groups, master data and authorised corrections in Django Admin.

4. Initial master-data setup

Complete master data before normal users begin entering assets. Open Administration from the navigation bar or browse to /admin/.

Step	Action	Expected result
1	Create Divisions.	Each division has a unique code and name.
2	Create Sections and Sub-sections under the correct parent.	Organisational hierarchy is available in forms and reports.
3	Create Locations with room numbers.	Assets can be assigned to physical locations.
4	Select “Is stock location” for stores/warehouse locations.	Return forms allow those locations as destinations.
5	Create Employees and associate them with division/section.	Employees can be selected as custodians and transfer recipients.
6	Create Suppliers.	Procurement forms can link supplier records.
7	Create Users and assign Groups.	Users see only permitted functions.

Data rule: Do not delete master records already used by assets. Mark them inactive where the model provides an active flag. Protected relationships prevent accidental loss of history.

5. Asset registration and maintenance

5.1 Register one asset manually

Step	Action	Expected result
1	Select Assets > Register asset.	The registration form opens.

Step	Action	Expected result
2	Complete Identification and Location. Asset Code and Transaction ID must be unique.	Core identification and current-state fields are ready.
3	Expand Procurement and enter quantity, amount, PO, supplier and bill details.	Acquisition information is linked to the asset.
4	Expand Stock and Installation and enter stock-entry reference, installation date, warranty months and status.	Installation details are linked to the asset.
5	Select Save asset.	The asset detail page opens and a QR token is available.

5.2 Edit an asset

Open the asset from the asset list and select Edit. Use editing for corrections to descriptive or procurement data. Do not directly change the current location or custodian to represent a real transfer; use the Transfer workflow so a movement record is created.

Audit rule: Every real-world movement must use Transfer, Return or Gate Pass. Editing the current-state fields directly should be restricted to authorised corrections.

5.3 Search and filter

The Assets page searches Asset Code, Transaction ID, description, item serial number, make and model. The status selector limits results to one current status. Search results update without a full page reload when the local HTMX-compatible script is enabled; ordinary form navigation remains available as a fallback.

5.4 Asset detail page

- Identification and current status.
- Current location and custodian.
- Procurement and installation summary.
- QR image.
- Recent movement history.
- Links to edit the record when the user has permission.

6. Transfers and returns

6.1 Transfer an asset

Step	Action	Expected result
1	Select Movements > Transfer.	The transfer form opens.

Step	Action	Expected result
2	Select the asset, destination location and recipient employee.	The intended new state is defined.
3	Enter voucher number, movement date and remarks.	The supporting reference is captured.
4	Select Execute transfer.	The service validates the asset, creates AssetMovement, and updates location/custodian in one database transaction.
5	Open the asset detail page.	The new state and movement history are visible.

6.2 Return an asset to stock

Step	Action	Expected result
1	Ensure the destination location is marked as a stock location in Administration.	The location appears in the Return form.
2	Select Movements > Return to stock.	The return form opens.
3	Select the asset, stock location and returning employee, then enter voucher/date/clause.	Return evidence is captured.
4	Select Record return.	A return movement is created, custodian is cleared and status becomes In stock.

Validation: Disposed, written-off and passed-out assets cannot be transferred. Returns must go to a location marked as stock.

7. Gate-pass and security workflow

7.1 Outward process

Step	Action	Expected result
1	Division Officer selects Gate Passes > Request gate pass.	A new request form opens.
2	Select Temporary or Permanent, division, purpose, destination and one or more assets.	The requested movement is defined.
3	For a Temporary gate pass, enter Expected return date.	The system can identify overdue returns.

Step	Action	Expected result
4	Submit request.	Status becomes Requested and a gate-pass number is generated.
5	Stores Officer opens the request and selects Stores approve.	Status becomes Stores approved.
6	Security Officer verifies the items and selects Mark outward.	Each asset receives a Gate outward movement and status becomes Outside.

7.2 Inward process for temporary gate passes

Step	Action	Expected result
1	Security Officer opens an outward temporary gate pass when the assets return.	The Mark inward action is available.
2	Verify the physical asset codes or QR pages.	Returned items match the gate-pass list.
3	Select Mark inward.	Each asset receives a Gate inward movement; its normal Assigned/In stock status is restored and the gate pass closes.

7.3 Permanent gate passes

A permanent gate pass closes after Security marks it outward. The asset remains Outside premises unless a later authorised process changes its status. Organisational policy should determine whether a permanent gate pass must also create a disposal or passout record.

8. QR-code workflow

Every asset receives a random, immutable QR token. The QR encodes only a URL; it does not embed employee or full asset data directly in the code.

Step	Action	Expected result
1	Open an asset and select QR PNG, or save/print the QR image shown on the detail page.	A PNG suitable for a label is produced.
2	Attach the label to the correct physical asset.	The code can be scanned by a phone or security scanner.
3	Scan the QR and sign in when QR_PAGE_PUBLIC=False.	The asset webpage shows code, description, serial number, current status/location/custodian, active gate-pass status and last movement.

Step	Action	Expected result
4	Compare the webpage with the physical item and gate pass.	Security can verify that the correct asset is moving.

Privacy: The secure default requires sign-in for QR pages. Set QR_PAGE_PUBLIC=True only after the organisation explicitly approves public visibility of the displayed fields.

9. Excel import

The package includes sample_data/Asset_Import_Template.xlsx. The importer uses a staging process: upload, validate, preview, confirm. It does not write valid rows into the live asset tables until confirmation.

9.1 Prepare the workbook

- Keep the header row unchanged.
- TRANSACTION_ID and BRIEF_DESCRIPTION are required.
- Use unique ASSET_CODE and TRANSACTION_ID values.
- Use YYYY-MM-DD for text dates, or real Excel date cells.
- Use numbers without currency symbols for amounts.
- Use YES or NO for transfer/return flags.
- Review organisational names for consistent spelling.

9.2 Upload and validate

Step	Action	Expected result
1	Select Import > Upload workbook.	The upload form opens.
2	Choose an .xlsx file up to 10 MB and add optional notes.	The file is ready for validation.
3	Select Upload and validate.	The system reads the first worksheet and creates staged ImportRow records.
4	Open the batch result.	Total, valid and invalid row counts and row-level errors are shown.
5	If errors exist, correct the original workbook and upload a new batch.	No live asset records are created from the invalid batch.
6	When all rows are valid, select Confirm import.	Assets and related procurement, installation, movement and disposal records are created atomically.

9.3 What the importer can create

- Divisions, sections, sub-sections, locations, employees and suppliers that do not already exist.
- Asset, Procurement and Installation records.
- Legacy transfer and return movements when the corresponding YES/NO fields are set.
- Disposal, auction, write-off or passout records when those fields contain values.

Control: Keep the original workbook and take a database backup before large imports. Test a small representative file in staging before importing the complete register.

10. Disposal, auction, write-off and passout

Step	Action	Expected result
1	Select Disposal > New record, or start from the relevant asset.	The proposal form opens.
2	Select the asset and enter lot name, disposal file number, auction ID, buyer, EMD, book value, financial year, sale values, write-off or passout references as applicable.	Commercial and approval-reference details are linked.
3	Select Create proposal.	The record is saved as Proposed and the asset becomes Under disposal.
4	An Approving Officer opens the record, reviews the evidence and changes the status when authorised.	Approved, Auctioned, Written off or Passed out status is applied.
5	Review the asset detail and disposal list.	The final state is visible and protected from ordinary transfer.

Approval: Only a user with the disposal approval permission can change the workflow status. When status changes, the server records that signed-in user as approved_by; the form never allows selecting another approver.

11. Reports and exports

Dashboard cards: Total assets, Assigned, In stock, Outside, Under disposal and temporary gate passes awaiting return.

Recent activity: Audit events for create, update, transfer, return, import and gate-pass actions.

Reports page: Acquisition value, assets by status, top locations and overdue temporary gate passes.

Asset register CSV: A downloadable CSV containing identification, current state, PO, supplier, amount and remarks.

QR webpage: A live per-asset webpage generated from the QR token.

Reporting note: The starter report page is intentionally simple. Add organisation-specific statutory, financial and reconciliation reports only after their definitions and approval rules are documented.

12. Administration and deployment

12.1 Docker quick start

Step	Action	Expected result
1	Extract the ZIP to the deployment folder.	Project files are available.
2	Copy .env.example to .env.	Environment configuration is created.
3	Replace every example password and set a long random DJANGO_SECRET_KEY.	Secrets are no longer default values.
4	Set LOAD_DEMO_DATA=False for production.	Demo records are not created.
5	Run docker compose up -d --build.	MySQL and Django containers start, migrations run, static files collect and roles bootstrap.
6	Open http://server:8000/accounts/login/ for an initial internal test.	The application sign-in page is available.

12.2 IIS reverse proxy on Windows Server 2022

- Install IIS, URL Rewrite and Application Request Routing (ARR).
- Enable proxy support in ARR.
- Create an IIS site with the organisation hostname and TLS certificate.
- Adapt deployment/iis-web.config.example to proxy to <http://127.0.0.1:8000>.
- Allow the HTTP_X_FORWARDED_PROTO server variable when required by IIS configuration.
- Set DJANGO_ALLOWED_HOSTS and DJANGO_CSRF_TRUSTED_ORIGINS to the public hostname.
- After HTTPS works, set DJANGO_SECURE_COOKIES=True and restart the web container.
- Restrict port 8000 so it is reachable only from the local server/reverse proxy.

12.3 Environment settings

Setting	Purpose	Production guidance
DJANGO_SECRET_KEY	Cryptographic signing key	Use a long random secret; never commit .env.
DJANGO_DEBUG	Debug mode	False in production.

Setting	Purpose	Production guidance
DJANGO_ALLOWED_HOSTS	Accepted Host headers	List the real application hostnames.
DJANGO_CSRF_TRUSTED_ORIGINS	Trusted browser origins	Use full https:// origins.
DJANGO_SECURE_COOKIES	Secure session/CSRF cookies	True after HTTPS is active.
QR_PAGE_PUBLIC	Anonymous QR visibility	False unless explicitly approved.
MYSQL_*	Database connection and passwords	Use strong unique credentials and protect backups.
DJANGO_SUPERUSER_*	Initial administrator bootstrap	Change immediately; do not leave defaults.
LOAD_DEMO_DATA	Create sample records	False in production.

13. Backup, restore and maintenance

13.1 What must be backed up

- MySQL database volume.
- Media volume containing uploaded imports and attachments.
- .env and deployment configuration, stored securely and separately.
- Application source/version used for each deployment.

13.2 Backup procedure

Step	Action	Expected result
1	Run the database dump and media archive commands in deployment/backup.sh or equivalent PowerShell automation.	Timestamped backup files are created.
2	Copy backups to encrypted off-server storage.	A server failure does not destroy the only backup.
3	Record backup success/failure and retention.	Operations can prove backups are occurring.
4	Regularly restore into a non-production environment.	The organisation verifies that backups are usable.

13.3 Upgrade procedure

1. Take verified backups.
2. Review dependency and Django security updates.
3. Build and test a staging copy.
4. Run python manage.py check and python manage.py test.
5. Deploy with docker compose up -d --build.
6. Review logs and execute representative workflows.

14. Troubleshooting

Symptom	Likely cause	Action
Cannot sign in	Wrong credentials, inactive user or service unavailable	Confirm account in Administration; check docker compose ps and web logs.
Permission denied	User lacks the required group/permission	Assign the correct minimum group and sign in again.
Return location not listed	Location is not marked as stock	Edit the location in Administration and select Is stock location.
Excel batch has invalid rows	Missing required fields, date/amount format or duplicates	Read each row error, correct the source workbook and upload a new batch.
CSRF verification failed behind IIS	Trusted origin or forwarded protocol is wrong	Set the HTTPS origin and X-Forwarded-Proto handling, then restart.
Static styling missing	Static collection or container restart issue	Run collectstatic and restart the web service.
QR redirects to sign-in	Secure QR default is enabled	Sign in, or obtain formal approval before setting QR_PAGE_PUBLIC=True.
Database connection error	MySQL container unhealthy or credentials mismatch	Check db logs, .env values and MYSQL_HOST=db.

15. Go-live checklist

- ☐ Production hostnames, TLS certificate and IIS reverse proxy tested.
- ☐ Default secrets and passwords replaced.
- ☐ DEBUG disabled and secure cookies enabled.
- ☐ Demo data disabled or removed.
- ☐ Divisions, sections, locations, employees and suppliers verified.
- ☐ User groups and permissions reviewed by process owners.
- ☐ Representative assets registered manually and by Excel import.
- ☐ Transfer, return, temporary gate pass, permanent gate pass and disposal workflows accepted by users.
- ☐ QR labels tested with the intended scanner/device and network.
- ☐ CSV and operational reports reconciled against source data.
- ☐ Database and media backups scheduled, encrypted and restored successfully in a test environment.
- ☐ Security review, vulnerability scanning and change approval completed.
- ☐ Support ownership, incident process and upgrade window documented.

Appendix A. Source-field mapping

The following table maps the supplied field labels to the implemented model or workflow. Field names are preserved for import aliases and reporting where practical.

Business area	Supplied fields	Implemented location
Asset identification	SL_NO, TRANSACTION_ID, NC_NO, NC_DATE, BRIEF_DESCRIPTION, SPECIFICATION, MAKE, MODEL, ITEM_SL_NO	Asset
Procurement	QTY, AMOUNT, PO_NO, PO_DATE, SUPPLIER_NAME, BILL_NO, BILL_DATE, BILL_VALUE	Procurement
Stock and installation	STOCK_ENTRY_REFERENCE, DATE OF INSTALLATION, WARRANTY PERIOD, STATUS OF ASSET	Installation
Location and responsibility	ROOM_NO, ROOM_IN_CHARGE_NAME, EMP_ID, EMP_NAME, DIVISION, SECTION, SUB SECTION, MAIN LOCATION, CURRENT USER	Asset + Division/Section/SubSection/Location/Employee
Transfer	TRANSFER (YES/NO), TRANSFER_FROM_NAME, TRANSFER_FROM_ID, TRANSFER_TO_ID, TRANSFER_TO_NAME, TRANS_VOUCHER_NO, TRANSFER_VOUCHER_DATE	AssetMovement (TRANSFER)
Return	RETURN (YES/NO), RETURNED_STOCK, RETURN_FROM_NAME, RET_VOUCHER_NO, RETURN_VOUCHER_DATE, RETURN_CLAUSE, RETURN_FROM_DIVISION	AssetMovement (RETURN)
Disposal and auction	DISPOSAL_DETAILS_LOT NAME, DISPOSAL FILE NO, AUCTION ID, H1_BUYER, EMD AMOUNT, TOTAL BOOK VALUE, FINANCIAL YEAR, REALIZED SALE VALUE, APPORTIONED SALE VALUE	DisposalRecord
Write-off and passout	WRITE OFF OM NO, WRITE OFF OM DATE, PASSOUT_FOR, PASSOUT_TYPE, PASSOUT_NO, PASSOUT_DATE	DisposalRecord
General	REMARKS	Asset/transaction remarks as appropriate

Appendix B. Workflow summary

Asset lifecycle

Start -> Asset Identification -> Procurement -> Stock and Installation -> Location Assignment -> Transfer? -> Return? -> Disposal? -> Write-off/Passout -> Remarks.

Gate-pass outward

Request from Division -> Temporary or Permanent selection -> Stores approval -> Security outward marking.

Gate-pass inward

Security verifies/scans returning temporary assets -> Inward marking -> Asset status restored -> Gate pass closed.

Excel import

Upload workbook -> Validate rows -> Preview errors -> Confirm -> Create normalised records.

QR reporting

Scan QR URL -> Authenticate (secure default) -> View live asset and gate-pass status.

Implementation notes

The source package contains README.md, docker-compose.yml, Dockerfile, committed migrations, tests, the Excel import template, an IIS configuration example, backup scripts and the Markdown administrator guide. Organisational process owners should approve any changes to workflow states, mandatory fields, reports or permissions before development changes are deployed.